

Investigating a raised CK

(in the presence or absence of symptoms)

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Red Whale

GEMS

Guidelines & Evidence Made Simple

If **NEW** muscle aches or raised CK in the context of starting or up-titrating a statin, go to the **Lipids and statins GEMS** and follow section 5 (statin intolerance pathway).

Follow the flow diagram below if the symptoms are **NOT** thought to be statin related.

Check CK if any significant symptoms: muscle pain or tenderness or weakness.

This flow diagram can also be used if CK is checked for whatever reason (symptoms or not) and is raised.

Other tests: also check renal function and urine sample (dipstick and M,C&S).

Look for precipitating causes, especially drugs, e.g. statin (see list of causes at the bottom of the page).

ULN = upper limit of normal (CK varies from lab to lab and between the sexes).

Admission needed?

CK <10x ULN
(approx.
<1000 IU/L)
and well:
manage in
primary care

CK 10–50x ULN (approx. 1000–5000 IU/L)

Stop any causative drugs

(if on statin, follow statin intolerance pathway (section 5 of
Lipids and statin GEMS).

**Discuss with medics and admit if AKI, oliguria or myoglobinuria (seen in
rhabdomyolysis)**

(myoglobinuria presents as dark urine, often with ≥2+ blood on dipstick but no
red cells on microscopy – but do not wait for microscopy results if suspected!).

**If severe/continuous muscle aches, consider autoimmune statin-related
necrotising myositis:** extremely rare (1–2/million treated), but very significant. It
causes severe progressive muscle weakness, even when statins are stopped.

CK >50x ULN
(approx.
>5000 IU/L)

**ADMIT SAME
DAY. Stop any
causative drugs**

Management in primary care

Assess symptoms and look for underlying cause(s) – see below.

CK <4x ULN (approx. <500 IU/L)

**Usually incidental/due to transient cause, e.g.
exercise, and will settle spontaneously.**

If no/mild symptoms and recent history of high-intensity
exercise, advise rest and good hydration, and recheck
CK in 1–2w (CK will usually normalise in 3–7d).

Review lifestyle factors, e.g. alcohol, medication and
personal/family history of muscle disease.
A recurrently stable CK <500 IU/L with no symptoms or
risk factors for an underlying cause needs no further
investigation.

CK >4 to <10x ULN

More likely to be due to a secondary cause.

Repeat CK and assess for any other causative drugs.
Arrange bloods, including U&Es, LFTs, TFTs, corrected
calcium, phosphate, magnesium and vitamin D. Check
urine dipstick.

Consider other endocrine investigations/inflammatory
markers if warranted based on history/comorbidities.

Autoimmune causes are commoner in those >50y with
symptoms. Rheumatology referral may be of benefit.

Causes of a raised CK

Most commonly physiological, due to exercise (+/- dehydration) or drugs. May be multifactorial:

Muscle insult: exercise (e.g. resistance training or extreme exercise), trauma, seizures, surgery, systemic illness.

Substance misuse: alcohol excess (and thiamine deficiency), stimulants, e.g. cocaine, MDMA.

Drugs: statins/fibrates (especially if other drugs affecting CYP3A4), antipsychotics, hydroxychloroquine,
colchicine, systemic retinoids, antiretrovirals for HIV, some cancer drugs.

Endocrine: thyroid disease, hyperparathyroidism, Cushing's, acromegaly.

Autoimmune: rheumatoid arthritis, SLE, coeliac disease, dermatomyositis, polymyositis.

Hereditary myopathies: e.g. glycogen storage disease, muscular dystrophy.

We make every effort to ensure the information in these pages is accurate and correct at the date of publication, but it is of necessity of a brief and general nature. The information presented herein should not replace your own good clinical judgement, or be regarded as a substitute for taking professional advice in appropriate circumstances. In particular, we suggest you carefully consider the specific facts, circumstances and medical history of any patient, and recommendations of the relevant regulatory authorities. We also suggest that you check drug doses, potential side-effects and interactions with the British National Formulary. Save insofar as any such liability cannot be excluded at law, we do not accept any liability for loss of any type caused by reliance on the information in these pages. December 2023. For full references see the relevant Red Whale articles.